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GUIDANCE TOOLS, SYSTEMS, AND METHODS

Abstract

A guidance tool includes a body having a length extending from a first end to a second end. The body includes a shape memory section along the length of the body. The shape memory section has a curved shape.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS AND INCORPORATION BY REFERENCE [0001] This application is a divisional filed under 37 C.F.R. § 1.53 claiming the benefit under 35 U.S.C. § 120 of any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application, including U.S. patent application Ser. No. 17/776,893, filed May 13, 2022, International Patent Application No. PCT/US2020/065189, filed Dec. 16, 2020, and U.S. Provisional Patent Application No. 62/962,610, filed Jan. 17, 2020, and are hereby incorporated by reference in accordance with 37 C.F.R. §§ 1.57; 1.97; and 1.98 in their entireties This application also incorporates by reference the entire disclosures of commonly assigned U.S. Pat. No. 8,808,303, entitled “Orthopedic Surgical Guide;” U.S. Pat. No. 9,675,365, entitled “System and Method for Anterior Approach for Installing Tibial Stem;” and U.S. Pat. No. 10,456,179, entitled “Intramedullary Ankle Technique and System.”.

FIELD OF DISCLOSURE

[0002] The disclosed systems and methods relate to surgical tools and methods. More particularly, the disclosed systems and methods relate to guiding a flexible reamer, including flexible reamer that may be used to form an intramedullary canal.

BACKGROUND

[0003] Many surgical procedures use rotating cutting tools, such as reamers, to form cavities or channels within bone. One example of such a surgical procedure is a total ankle replacement (“TAR”) procedure in which an intramedullary channel may be formed in a tibia so that the tibia may receive a stem component. The intramedullary channel typically is formed along the mechanical axis of the tibia, and many conventional techniques require the violation of additional bones beyond the tibia (e.g., the talus and calcaneus) to form the intramedullary channel. One example of such a technique is disclosed in commonly assigned U.S. Pat. No. 8,808,303, which has been incorporated by reference above. Violating additional bones beyond the tibia may increase the length of the surgery and risk of infection or other complications.

SUMMARY

[0004] In some embodiments, a guidance tool includes a body having a length extending from a first end to a second end. The body includes a shape memory section along the length of the body. The shape memory section has a curved shape.

[0005] In some embodiments, a method includes inserting a first end of a guidance tool into an end of a bone; advancing the guidance tool into the bone until a shape memory section of the guidance tool is disposed adjacent to the end of the bone; cutting the guidance tool at a location between the shape memory section and a second end of the guidance tool; and advancing a cutting tool along the guidance tool to form a cavity in the bone. Cutting the guidance tool allows the shape memory section of the guidance tool to regain its programmed shape.

[0006] In some embodiments, a method includes coupling a fixture along a length of a bone such that a hole defined by the fixture is positioned adjacent to a side of the bone; inserting a guidance tool through the hole of the fixture and into the bone until the guidance tool extends along a length of the bone and a leading end of the guidance tool is exposed adjacent to an end of the bone; and advancing a cutting tool along the guidance tool to form a cavity in the bone.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** illustrates one example of a human foot;

[0008] FIG. **2** illustrates one example of a resected tibia in accordance with some embodiments;

[0009] FIG. **3** illustrates one example of a resected tibia and a resected talus in accordance with some embodiments;

[0010] FIG. **4** is a frontal plane view of one example of a guidance tool being inserted into a tibia in accordance with some embodiments;

[0011] FIG. **5** is a sagittal plane view of a cutting tool being guided by the guidance tool shown in FIG. **4** in accordance with some embodiments;

[0012] FIG. **6** is a frontal plane view of one example of a fixture and guidance tool in accordance with some embodiments; and

[0013] FIG. **7** is a sagittal plane view of the fixture illustrated in FIG. **6** in accordance with some embodiments.

DETAILED DESCRIPTION

[0014] This description of preferred embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. The drawing figures are not necessarily to scale and certain features may be shown exaggerated in scale or in somewhat schematic form in the interest of clarity and conciseness. In the description, relative terms such as “horizontal,” “vertical,” “up,” “down,” “top” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing figure under discussion. These relative terms are for convenience of description and normally are not intended to require a particular orientation. Terms including “inwardly” versus “outwardly,” “longitudinal” versus “lateral” and the like are to be interpreted relative to one another or relative to an axis of elongation, or an axis or center of rotation, as appropriate. Terms concerning attachments, coupling and the like, such as “connected” and “interconnected,” refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. When only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein. The term “operatively connected” is such an attachment, coupling or connection that allows the pertinent structures to operate as intended by virtue of that relationship. In the claims, means-plus-function clauses, if used, are intended to cover the structures described, suggested, or rendered obvious by the written description or drawings for performing the recited function, including not only structural equivalents but also equivalent structures.

[0015] The disclosed systems and methods advantageously facilitate the intramedullary guidance while minimizing and/or eliminating the violation of adjacent bones as is typically done during conventional surgical procedures. While the systems and methods are described in connection with performing TAR, one of ordinary skill in the art will understand that the disclosed systems and methods may be used to facilitate the creation of intramedullary canals or channels in other bones or body parts.

[0016] FIG. **1** illustrates one example of a human foot **10** and ankle **12**. As is known, the human foot **10** includes a number of bones, including the talus **14**, which sits atop the calcaneus **20**. The talus **14** forms part of the ankle joint with the tibia **16**, which is positioned adjacent to the fibula **18**. To prepare an ankle **12** for a TAR, the talus **14** and tibia **16** may be resected to provide a resected joint space **22** as best seen in FIGS. **2** and **3**. Examples of tools and procedures for forming the resected joint space **22** are disclosed in U.S. Pat. Nos. 8,808,303 and 9,675,365, which have been incorporated by reference above.

[0017] FIGS. **4** and **5** illustrate one example of a guidance tool in accordance with some embodiments. Guidance tool **100** has an elongate shape extending from a first end **102**, which may be an insertion end, to a second end **104**, which may be a trailing end. End **102** may include a point or taper for facilitating insertion of the guidance tool **100** into a medium, such as skin or bone. Guidance tool **100** may be formed from shape memory material, such as a shape memory alloy

selected from the group consisting of Cu—Al—Ni, NiTi (e.g., Nitinol), Fe—Mn—Si, Cu—Zn—Al, and Cu—Al—Ni.
[0018] In some embodiments, guidance tool **100** may include a centering mechanism **108** for centering the guidance tool within a bone. For example, the centering mechanism **108** may include a balloon or stent material configured to expand from a first, collapsed configuration to a second, expanded configuration. The guidance tool **100** is inserted into the medium (e.g., an intramedullary space, such as cancellous bone) with the centering mechanism **108** in its collapsed configuration. When guidance tool **100** has been inserted to its desired depth or location, the centering mechanism **108** may be deployed into its expanded configuration.
[0019] As will be understood by one of ordinary skill in the art, the manner in which the centering mechanism **108** is deployed may vary. For example, in embodiments where the centering mechanism **108** includes a balloon, then a gas, gel, or liquid may be injected into the internal chamber defined by the balloon to increase the size of the balloon. Techniques similar to those used in vertebroplasty and kyphoplasty may be used to expand the balloon. One of ordinary skill in the art will understand that as the diameter of the balloon increases the cancellous bone is compressed against the stronger cortical bone thereby centering the guidance tool within the bone. Additional materials, such as a stent material, may be provided along with a balloon material to guide the expansion and shape of the balloon.
[0020] In some embodiments, guidance tool **100** includes a stop **110** along its length. For example, stop **110** may take the form of a bead, taper, or other protrusion having an enlarged diameter relative to one or more portions of guidance tool **100** that are adjacent to stop **110**. Stop **110** may be disposed adjacent to centering mechanism **108** and is configured to stop the advancement of a cutting tool along the guidance tool **100**, as will be described in greater detail below, thereby protecting the centering mechanism **108** from being damaged by the cutting tool.
[0021] Guidance tool **100** may include a shape memory section or portion **112**. In some embodiments, the shape memory section **112** is curved to facilitate the guidance of a flexible cutting tool into an intramedullary canal. As described in greater detail below, the shape memory section **112** may be positioned along the length of the guidance tool **100** such that when the stop **110** is located at the desired location within the bone the shape memory section will be adjacent to an end of the bone.
[0022] In some embodiments, the guidance tool **100** may be placed within an inferior portion of a tibia **16** with the aid of a cannula. The tibia **16** and talus **14** may be prepared by making bony cuts, such as those described in U.S. Pat. No. 9,675,365, entitled “System and Method for Anterior Approach for Installing Tibial Stem,” which is incorporated by reference herein in its entirety. Once the inferior portion of the tibia **16** is resected to form a resected joint space **22**, the cannula may be positioned adjacent to the calcaneus **20**.
[0023] With the cannula positioned, the leading end **102** of guidance tool **100** is inserted into the cannula and advanced into the calcaneus **20**, talus **14**, and tibia **16**. The position of the guidance tool **100** within the tibia may be checked using fluoroscopy as will be understood by one of ordinary skill in the art. In some embodiments, the guidance tool **100** may include one or more markings (not shown) along its length. Each marking may identify a distance from the respective marking to the leading end **102** of the guidance tool **100** or a distance from the respective marking to the stop **110** such that a surgeon or physician will be able to determine if the guidance tool **100** has been inserted to the desired depth without the use of fluoroscopy.
[0024] In embodiments in which the guidance tool **100** is configured with a centering mechanism **108**, the centering mechanism may be activated or deployed to center the guidance tool **100** within the tibia **16**. For example, if the centering mechanism **108** includes a balloon that may be inflated around the guidance tool **100**, then the balloon is inflated, such as by injecting a gas, gel, or liquid into the internal chamber defined by the balloon. As the balloon is filled, the balloon expands, which results in the cancellous being compressed. The cancellous bone is compressed to the

stronger cortical bone which is sufficiently strong to withstand the pressure exerted by the balloon being inflated.

[0025] With the guidance tool **100** positioned within the tibia **16**, the shape memory section **112** may be deployed. In some embodiments, the shape memory section **112** is deployed by cutting the guidance tool **100**, such as by using a pin cutter, adjacent to the shape memory section **112** to form a cut end **114**. The cutting of the guidance tool **100** eliminates the straightening force that was being applied to the guidance tool **100** by the cannula such that the shape memory section **112** is now free to revert back to its programmed shape, which may be a curved shape. In some embodiments, the curved shape of the shape memory section **112** extends through the anterior window of the resected joint space **22**. In some embodiments, the shape memory section **112** may be deployed prior to positioning the cannula and activating or deploying the centering mechanism as will be understood by one of ordinary skill in the art.

[0026] The guidance tool **100** is now ready to guide a cutting tool, such as a flexible reamer **500**, to prepare an intramedullary channel for receiving an implant. In some embodiments, a flexible reamer, such as the flexible reamer **500** described in U.S. Pat. No. 10,456,179, entitled “Intramedullary Ankle Technique and System,” the entirety of which is incorporated by reference herein, is modified to provide a cannulated flexible reamer. The cannulated flexible reamer **500** defines a passageway **504** that extends through the nose **502** and the rest of the reamer **500**. In some embodiments, the body **506** of the cannulated flexible reamer **500** includes a plurality of segments. The segments **507-1**, **507-2**, . . . **507-***n* (collectively, “segments **507**”) may be movable (e.g., rotatable and/or pivotable) relative to an adjacent segment **507**. In some embodiment, each segment **507** is movable (e.g., rotatable and/or pivotable) relative to an adjacent segment such that the reamer **500** may bend along its length.

[0027] The cannulated flexible reamer **500** is inserted over the guidance tool **100**. More particularly, the leading end or nose **502** of the cannulated flexible reamer **500**, which may include one or more flutes or cutting surfaces **503**, is slid onto the cut end **114** of guidance tool **100** such that the guidance tool **100** is received within passageway **504** of reamer **500** as shown in FIG. **5**. The flexible reamer **500** is advanced along the guidance tool **100**, including along the shape memory section **112**, and up into the tibia **16** to form an intramedullary channel. The flexible reamer is advanced along the length of the guidance tool **100** until it contacts stop **110** or till a desired depth has been achieved. In some embodiments, the guidance tool **100** may include markings along its length to provide a visual aid to the surgeon.

[0028] Once the intramedullary channel has been formed, the cannulated flexible reamer **500** may be slid off the guidance tool **100**. The guidance tool **100** may then be removed from the tibia **16**. In some embodiments in which the guidance tool **100** includes a centering mechanism **108** in the form of a balloon, the balloon may be deflated prior to removing the guidance tool **100** from its placement within the tibia **16**.

[0029] FIGS. **6** and **7** illustrate another example of the insertion of a guidance tool into a bone canal in accordance with some embodiments. As shown in FIGS. **6** and **7**, the guidance tool **100** is inserted into the bone canal with the assistance of a fixture **200**. Fixture **200** may include a body **202** that may be coupled to the bone or to an extramedullary guidance device, such as, for example, a foot holder and alignment tool **300** described in U.S. Pat. No. 8,808,303, as will be understood by one of ordinary skill in the art.

[0030] In some embodiments, body **202** is sized and configured such that the length of the body **202** extends across or substantially across a width of a bone, such as a human tibia **16**. Body **202** of fixture **200** may include a flange **204** extending at an angle (e.g., perpendicularly) with respect to a longitudinal axis defined by the body **202**. Body **202** defines at least one hole **206** sized and configured to receive a guidance tool, such as guidance tool **100** described above or a conventional k-wire. In some embodiments, hole **206** is configured to allow the guidance tool to be inserted at a non-orthogonal angle relative to an axis defined by the bone (e.g., a mechanical axis or a longitudinal axis). Body **202** may also include one or more gunsights **208** for providing an alignment check.

[0031] Body **202** may also include one or more adjustment mechanisms, including an anterior-posterior ("AP") adjustment block **210**, a medial-lateral ("ML") adjustment block **212**, and a proximal-distal adjustment block **214**. The implementation of the adjustment mechanisms may be varied. For example, in some embodiments, the adjustment mechanisms include a threaded thumb screw **216** having an enlarged foot **218** that is disposed at an opposite end of thumb wheel **220**. In some embodiments, the foot **218** is sized and configured to contact, but not cause damage or dig into, a bone, body **202** of fixture **200**, or the extramedullary guidance tool **100** as will be understood by one of ordinary skill in the art.

[0032] In use, the fixture **200** is positioned along a length of a bone, such as a tibia **16**, after the bone has been prepared to accept a guidance tool **100**. For example, in embodiments in which the fixture **200** and guidance tool **100** are to be used to guide a flexible reamer, e.g., flexible reamer **500**, to form an intramedullary channel for receiving a tibial component of an ankle prosthesis, the inferior portion **16***a* of the tibia **16** is resected to form a resected joint space **22** as shown in FIGS. **2** and **3**. The resected joint space may be formed as described in U.S. Pat. No. 9,675,365, entitled "System and Method for Anterior Approach for Installing Tibial Stem," which has been incorporated by reference above.

[0033] The fixture **200** may be positioned along the length of the bone by using a strap (not shown) to couple the fixture **200** directly to the bone or by coupling the fixture **200** to an external guidance device (also not shown). Positioning of the fixture **200** along the length of the bone may also include adjusting the location of the fixture **200** using one or more of the adjustment mechanisms, i.e., AP adjustment block **210**, ML adjustment block **212**, and proximal-distal adjustment block **214**. The adjustment mechanisms may be used by rotating the respective threaded thumb screw **216** as will be understood by one of ordinary skill in the art. The alignment of the fixture **200** relative to the bone may be checked using gunsights **208**, which may include visualizing the fixture and bone using fluoroscopy as will be understood by one of ordinary skill in the art.

[0034] Once the fixture **200** is positioned, a guidance tool **100** may be inserted into the hole **206** of fixture **200**. Hole **206** may be positioned along a medial or lateral side of the bone such that the guidance tool **100** is inserted into a medial or lateral aspect of the bone. The guidance tool **100** is routed through fixture **200** and along the length of the bone until a leading end extends from the inferior portion of the tibia **16** into resected joint space **22**. For example, in some embodiments, the guidance tool **100** is pre-bent or has sufficient flexibility such that when the guidance tool **100** is inserted through the fixture at an angle it may be manipulated by the surgeon until the desired orientation within the bone is achieved.

[0035] With the leading end of the guidance tool **100** exposed within the resected joint space **22**, a cutting tool, such as a flexible reamer, is used to prepare an intramedullary channel for receiving an implant. As described above, a flexible reamer, such as the flexible reamer described in U.S. Pat. No. 10,456,179, entitled "Intramedullary Ankle Technique and System" and which has been incorporated by reference above, is modified to provide a cannulated flexible reamer that is inserted over the leading end of guidance tool. The flexible reamer is advanced along the guidance tool **100** and up into the tibia **16** to form an intramedullary channel.

[0036] Once the intramedullary channel has been formed, the cannulated flexible reamer may be slid off of the guidance tool **100**. The guidance tool **100** may then be removed from the tibia **16**. Fixture **200** may then also be removed from its engagement with the bone.

[0037] In some embodiments, a guidance tool includes a body having a length extending from a first end to a second end. The body includes a shape memory section along the length of the body. The shape memory section has a curved shape.

[0038] In some embodiments, a stop is positioned along the length of the body adjacent to the first end. The stop has an enlarged dimension relative to portions of the body that are directly adjacent

to the stop.
[0039] In some embodiments, the body includes a centering mechanism disposed between the stop and the first end of the body.
[0040] In some embodiments, the centering mechanism includes an expandable portion configured to expand from a collapsed configuration to an expanded configuration.
[0041] In some embodiments, the centering mechanism includes a balloon.
[0042] In some embodiments, the shape memory section is positioned between the stop and the second end of the body.
[0043] In some embodiments, the body is formed from a shape memory alloy.
[0044] In some embodiments, the shape memory alloy is selected from the group consisting of Cu—Al—Ni, NiTi, Fe—Mn—Si, Cu—Zn—Al, and Cu—Al—Ni.
[0045] In some embodiments, a method includes inserting a first end of a guidance tool into an end of a bone; advancing the guidance tool into the bone until a shape memory section of the guidance tool is disposed adjacent to the end of the bone; cutting the guidance tool at a location between the shape memory section and a second end of the guidance tool; and advancing a cutting tool along the guidance tool to form a cavity in the bone. Cutting the guidance tool allows the shape memory section of the guidance tool to regain its programmed shape.
[0046] In some embodiments, the method includes expanding a centering mechanism of the guidance tool to center the guidance tool within the bone prior to cutting the guidance tool.
[0047] In some embodiments, expanding the centering mechanism includes inflating a balloon.
[0048] In some embodiments, the cutting tool includes a cannulated flexible reamer.
[0049] In some embodiments, the cutting tool is advanced along the guidance tool until the cutting tool contacts a stop disposed adjacent to a first end of the guidance tool.
[0050] In some embodiments, the method includes resecting an end of the bone to form a resected joint space between the bone and a second bone prior to inserting the end of the guidance tool into the end of the bone.
[0051] In some embodiments, a cut end of the guidance tool extends through a window of the resected joint space when the shape memory section of the guidance tool regains its programmed shape.
[0052] In some embodiments, the bone is a tibia.
[0053] In some embodiments, a method includes coupling a fixture along a length of a bone such that a hole defined by the fixture is positioned adjacent to a side of the bone; inserting a guidance tool through the hole of the fixture and into the bone until the guidance tool extends along a length of the bone and a leading end of the guidance tool is exposed adjacent to an end of the bone; and advancing a cutting tool along the guidance tool to form a cavity in the bone.
[0054] In some embodiments, the method includes forming a resected joint space between the bone and a second bone prior to inserting the guidance tool into the bone.
[0055] In some embodiments, the bone is a tibia.
[0056] In some embodiments, the cutting tool is a cannulated flexible reamer.
[0057] In some embodiments a system includes a guidance tool and a cannulated flexible reamer. The guidance tool has a body with a length extending from a first end to a second end. The body includes a shape memory section along the length of the body. The shape memory section has a curved shape. The cannulated flexible reamer defining a channel sized and configured to receive at least a portion of the guidance tool. A leading end of the cannulated flexible reamer includes plurality of flutes.
[0058] In some embodiments, a body of the cannulated flexible reamer includes a plurality of segments. At least one of the segments is movable relative to an adjacent segment. In some embodiments, each segment is movable relative to an adjacent segment.
[0059] In some embodiments a system includes a guidance tool, a cannulated flexible reamer, and a fixture. The guidance tool has a body with a length extending from a first end to a second end. The

body includes a shape memory section along the length of the body. The shape memory section has a curved shape. The cannulated flexible reamer defining a channel sized and configured to receive at least a portion of the guidance tool. A leading end of the cannulated flexible reamer includes plurality of flutes. The fixture is sized and configured to be disposed along a length of a bone. The fixture defines a hole that is positioned along the length of fixture such that, when the fixture is disposed adjacent to a bone, the guidance tool may be received in the hole of the fixture and into the bone.
[0060] In some embodiments, a body of the cannulated flexible reamer includes a plurality of segments. At least one of the segments is movable relative to an adjacent segment. In some embodiments, each segment is movable relative to an adjacent segment.
[0061] Although the invention has been described in terms of exemplary embodiments, it is not limited thereto. Rather, the appended claims should be construed broadly, to include other variants and embodiments of the invention, which may be made by those skilled in the art without departing from the scope and range of equivalents of the invention.

Claims

1. A method, comprising: inserting a first end of a guidance tool into an end of a bone; advancing the guidance tool into the bone until a shape memory section of the guidance tool is disposed adjacent to the end of the bone; cutting the guidance tool at a location between the shape memory section and a second end of the guidance tool, wherein cutting the guidance tool causes the shape memory section of the guidance tool to regain its programmed shape; and advancing a cutting tool along the guidance tool to form a cavity in the bone.
2. The method of claim 1, further comprising an expanding centering mechanism of the guidance tool to center the guidance tool within the bone.
3. The method of claim 2, wherein expanding the centering mechanism includes inflating a balloon.
4. The method of claim 1, wherein the cutting tool includes a cannulated flexible reamer.
5. The method of claim 1, wherein the cutting tool is advanced along the guidance tool until the cutting tool contacts a stop disposed adjacent to a first end of the guidance tool.
6. The method of claim 1, further comprising resecting an end of the bone to form a resected joint space between the bone and a second bone prior to inserting the end of the guidance tool into the end of the bone.
7. The method of claim 6, wherein a cut end of the guidance tool extends through a window of the resected joint space when the shape memory section of the guidance tool regains its programmed shape.
8. The method of claim 1, wherein the bone is a tibia.
9. A method comprising: coupling a fixture along a length of a bone such that a hole defined by the fixture is positioned adjacent to a side of the bone; inserting a guidance tool through the hole of the fixture and into the bone until the guidance tool extends along a length of the bone and a leading end of the guidance tool is exposed adjacent to an end of the bone; and advancing a cutting tool along the guidance tool to form a cavity in the bone.
10. The method of claim 9, further comprising forming a resected joint space between the bone and a second bone prior to inserting the guidance tool into the bone.
11. The method of claim 10, wherein the bone is a tibia.
12. The method of claim 9, wherein the cutting tool is a cannulated flexible reamer.
13. A method comprising: coupling a fixture along a length of a bone such that a hole defined by the fixture is positioned adjacent to a side of the bone; providing a guidance tool, including a body having a length extending from a first end to a second end, the body including a shape memory section along the length of the body, the shape memory section having a curved shape, a centering mechanism disposed between the stop and the first end of the body, an expandable portion

configured to expand from a collapsed configuration to an expanded configuration, a stop is positioned along the length of the body adjacent to the first end, the stop having an enlarged dimension relative to portions of the body that are directly adjacent to the stop, wherein the shape memory section is positioned between the stop and the second end of the body; inserting a guidance tool through the hole of the fixture and into the bone until the guidance tool extends along a length of the bone and a leading end of the guidance tool is exposed adjacent to an end of the bone; and advancing a cutting tool along the guidance tool to form a cavity in the bone.

14. The method of claim 13, expanding the centering mechanism of the guidance tool to center the guidance tool within the bone.

15. The method of claim 14, wherein expanding the centering mechanism includes inflating a balloon.

16. The method of claim 13, wherein the cutting tool includes a cannulated flexible reamer.

17. The method of claim 13, wherein the cutting tool is advanced along the guidance tool until the cutting tool contacts the stop disposed adjacent to the first end of the guidance tool.

18. The method of claim 13, further comprising resecting an end of the bone to form a resected joint space between the bone and a second bone prior to inserting the end of the guidance tool adjacent to the end of the bone.

19. The method of claim 18, wherein a cut end of the guidance tool extends through a window of the resected joint space when the shape memory section of the guidance tool regains its programmed shape.

20. The method of claim 13, wherein the bone is a tibia.
